

DSA0502 QUERY PROCESSING FOR DATASCIENCE

CO1-AT2-APPLICATION-BASED QUESTIONS

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33. Weather Data Processing

Daily weather information is stored in weather.csv.

Task:

- Find the highest and lowest temperatures.
- Calculate the average rainfall.
- Identify the days where rainfall exceeded 100 mm.
- Export the filtered data.

Aim

To analyze weather data from a CSV file using Python Pandas.

Procedure

1. Create a file named **weather.csv**.
2. Import the Pandas library.
3. Read the CSV file.
4. Find the highest and lowest temperatures.
5. Calculate the average rainfall.
6. Display days with rainfall above **100 mm**. 7. Save the filtered data into a new CSV file.

Program

```
import pandas as pd
df = pd.read_csv("weather.csv")

print("Highest Temperature:",
df["Temperature"].max())
print("Lowest Temperature:",
df["Temperature"].min())
print("Average Rainfall:",
```

```

df["Rainfall"].mean()) heavy_rain = df[df["Rainfall"] >
100] print("\nDays with Rainfall > 100 mm:")
print(heavy_rain)
heavy_rain.to_csv("heavy_rainfall_days.csv",
index=False)

```

Output

```

== RESTART: C:/Users/sindh/OneDrive/Desktop/Query/Experiments/CO1 AT2 CODE.py ==
Highest Temperature: 39
Lowest Temperature: 18
Average Rainfall: 74.25

Days with Rainfall > 100 mm:
   Date  Temperature  Rainfall
3  04-07-2026         35       110
5  06-07-2026         27       120
8  09-07-2026         25       145
10 11-07-2026         26       105
12 13-07-2026         24       130
15 16-07-2026         22       160
18 19-07-2026         27       115

Filtered data exported as 'heavy_rainfall_days.csv'
>>>

```

Result

The weather data was processed successfully. The highest and lowest temperatures, average rainfall, and days with rainfall above **100 mm** were identified. The filtered data was saved as **heavy_rainfall_days.csv**.

15. Social Media User Data (JSON)

User information is stored in `users.json`.

Task:

- **Display all usernames.**
- **Count verified users.**
- **Find users with more than 10,000 followers.**
- **Export the filtered records.**

Aim

To process user data from a JSON file using Python by displaying usernames, counting verified users, finding users with more than 10,000 followers, and exporting the filtered data.

Procedure

1. Create a file named **`users.json`**.
2. Import the **`json`** module.
3. Read the JSON file.
4. Display all usernames.
5. Count verified users.
6. Find users with more than **10,000 followers**.
7. Export the filtered records to a new JSON file.

Program:

```
import json

with open("users.json", "r") as file:

    users = json.load(file)

print("Usernames:")

for user in users:
```

```

print(user["username"])

verified = [user for user in users if user["verified"]]

print("\nVerified Users:", len(verified))

popular_users = [user for user in users if user["followers"] > 10000]

print("\nUsers with more than 10000 followers:")

for user in popular_users:

    print(user)

with open("popular_users.json", "w") as file:

    json.dump(popular_users, file, indent=4)

print("\nFiltered data exported as 'popular_users.json'")

```

Output:

```

>>> == RESTART: C:/Users/sindh/OneDrive/Desktop/Query/Experiments/CO1 AT2 CODE2.py =
Usernames:
john123
alice
david
emma
mike

Verified Users: 3

Users with more than 10000 followers:
{'username': 'john123', 'verified': True, 'followers': 12000}
{'username': 'david', 'verified': True, 'followers': 15000}

Filtered data exported as 'popular_users.json'

```

Result

The JSON user data was processed successfully. All usernames were displayed, verified users were counted, users with more than **10,000 followers** were identified, and the filtered records were exported to **popular_users.json**.